

DIRECTORATE OF ONLINE EDUCATION

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR – 603 203

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GEOSPATIAL DATA ANALYSIS USING MACHINE LEARNING

Submitted by

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Geospatial Data Analysis Using Machine Learning

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ABSTRACT

The project develops an intelligent, cloud-enabled Geo-AI analytics framework for environmental monitoring, vegetation assessment, and spatial risk prediction using Machine Learning and GIS technologies.

The system integrates satellite imagery (Sentinel-2, Landsat-8), OpenStreetMap vectors, Digital Elevation Models and ISRO-IIRS datasets to create a unified analytics platform.

Key spatial features extracted include NDVI, Land Surface Temperature (LST), elevation, rainfall, and land use/cover data for supervised ML tasks.

ML algorithms — Random Forest, Gradient Boosting, SVM, Decision Trees and CNNs — detect deforestation, urban expansion, vegetation stress and high-risk ecological zones.

Platform deployed as Flask + Streamlit cloud web app with JWT authentication, interactive Leaflet.js maps and Chart.js dashboards for real-time decision support.

INTRODUCTION

PROBLEM CONTEXT

- Geospatial analysis is critical for environmental monitoring, disaster management and climate analysis.
- Traditional systems rely on manual interpretation and static mapping — lacking intelligent prediction.
- AI & ML enable automatic pattern detection from large spatial datasets.
- Increasing satellite data availability demands scalable, automated analytics platforms.

PROPOSED SYSTEM

- Geo-AI Analytics Dashboard integrating ML + GIS for environmental risk zone prediction.
- Users upload CSV geospatial datasets and obtain predictions via interactive charts & maps.
- Platform inspired by ISRO-IIRS advanced space research & geospatial intelligence systems.
- Cloud-based, secure, scalable and user-friendly web interface.

OBJECTIVES & SCOPE

Main Objectives

1. Develop a system for analyzing geospatial data
2. Apply ML techniques for environmental prediction
3. Automate data preprocessing and feature engineering
4. Provide a user-friendly interactive web interface
5. Generate accurate and meaningful geospatial results

Project Scope

Includes:

- Structured CSV geospatial datasets
- ML-based classification & prediction
- Basic visualization of results
- Agriculture, environment, climate studies

Excludes:

- Real-time satellite data processing
- Advanced GIS software integration

EXISTING SYSTEM & PROBLEM DESCRIPTION

Lack of Intelligent Prediction

Traditional GIS platforms focus on visualization/mapping only — no ML-based predictive environmental analysis.

Poor Scalability

Cannot efficiently handle large satellite imagery, raster datasets and high-dimensional spatial information.

No Interactive Visualization

Systems lack dynamic dashboards and real-time mapping for effective spatial trend interpretation.

Limited ML-GIS Integration

Existing systems fail to combine spatial analytics with ML models to detect hidden environmental patterns.

Manual & Error-Prone Process

Significant manual intervention required for detecting changes like deforestation, urban expansion and climate risks.

No Real-Time Decision Support

Unable to generate real-time predictions needed for disaster management, agriculture and climate monitoring.

PROPOSED SOLUTION

ML Algorithms

Random Forest · Gradient Boosting · SVM · CNN · Decision Trees

Data Sources

Sentinel-2 · Landsat-8 · OpenStreetMap · DEM · ISRO-IIRS

Preprocessing

Georeferencing · Cloud masking · Normalization · NDVI · LST

Geo-AI Analytics Platform

Visualization

Leaflet.js maps · Chart.js · Interactive dashboards

Deployment

Flask · Streamlit · JWT Auth · REST API · Cloud-based

Applications

Precision Agriculture · Disaster Mgmt · Climate Monitoring · Urban Planning

SYSTEM REQUIREMENTS

Hardware Requirements

Component	Specification
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	20 GB Free Space
Internet	Required

Key Functional Requirements: User Registration & Login · CSV Dataset Upload · Environmental Risk Prediction · Data Visualization · Geospatial Map View · Admin Dashboard · JWT Authentication

Software Requirements

Component	Specification
Language	Python 3.x
IDE	Visual Studio Code
ML Libraries	Scikit-learn, TensorFlow/Keras
Web Framework	Flask, Streamlit
Frontend	HTML, CSS, JavaScript
Mapping	Leaflet.js, Chart.js
Data Handling	Pandas, NumPy
Database	JSON-based persistence

MODULE DESCRIPTION

01



Data Input Module

- Accepts CSV datasets with geospatial parameters
- Validates dataset format and structure
- Verifies latitude, longitude, rainfall, temperature, vegetation index
- Prevents invalid or corrupted data uploads

Python · Pandas · Flask/Streamlit upload components

03



Machine Learning Module

- Trains models on historical environmental data
- Implements classification & prediction algorithms
- Random Forest · Gradient Boosting · SVM · CNN
- Evaluates with Accuracy, F1-Score, RMSE, Confusion Matrix

Scikit-learn · TensorFlow/Keras · Python

05



Visualization Module

- Displays interactive geospatial Leaflet.js maps
- Generates Chart.js charts and statistical graphs
- Supports spatial risk mapping & prediction layers
- Visualizes prediction distributions & trend analysis

Leaflet.js · Chart.js · Matplotlib · Plotly

02



Data Preprocessing Module

- Handles missing & null values; removes duplicates
- Performs normalization, scaling & encoding
- Feature extraction: NDVI, LST, elevation, rainfall
- Splits data into training and testing sets

Pandas · NumPy · Scikit-learn preprocessing

04



Prediction Module

- Accepts new environmental input data
- Applies trained model for real-time inference
- Predicts risk levels & identifies vegetation stress zones
- Outputs risk prediction & classification results

joblib (model loading) · Flask API · Python

06



User Interface Module

- Responsive web dashboard with real-time display
- Secure JWT login & session management
- Dataset upload, Admin panel & prediction history
- Intuitive design requiring no technical expertise

HTML · CSS · JavaScript · Flask · Streamlit

MODULE DESCRIPTION — DATA FLOW & RESPONSIBILITIES



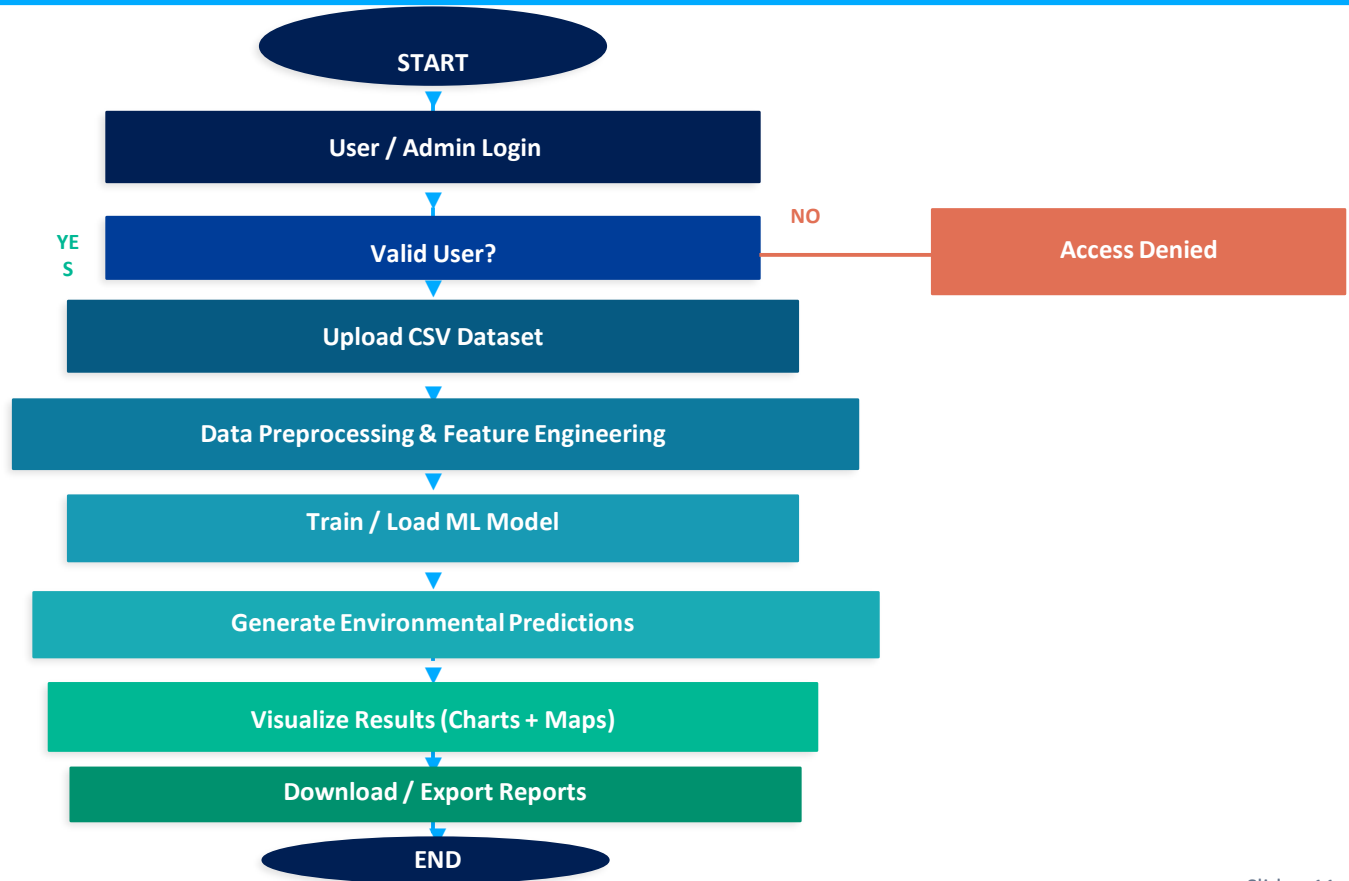
Module	Responsibility	Input → Output
Data Input Module	Accepts CSV · Validates attributes · Blocks corrupt files	Raw CSV file → Validated DataFrame
Preprocessing Module	Fill nulls · Normalize · Encode · Feature extract	Raw DataFrame → Clean feature matrix
ML Module	Train Random Forest · Evaluate · Save model.pkl	Feature matrix → Trained ML model

Module	Responsibility	Input → Output
Prediction Module	Load model · Run inference · Return risk labels	New CSV input → Prediction JSON
Visualization Module	Render Leaflet.js map · Chart.js graphs · Reports	Prediction results → Interactive dashboard
User Interface Module	JWT auth · File upload · Display results · Admin panel	User actions → Web dashboard response

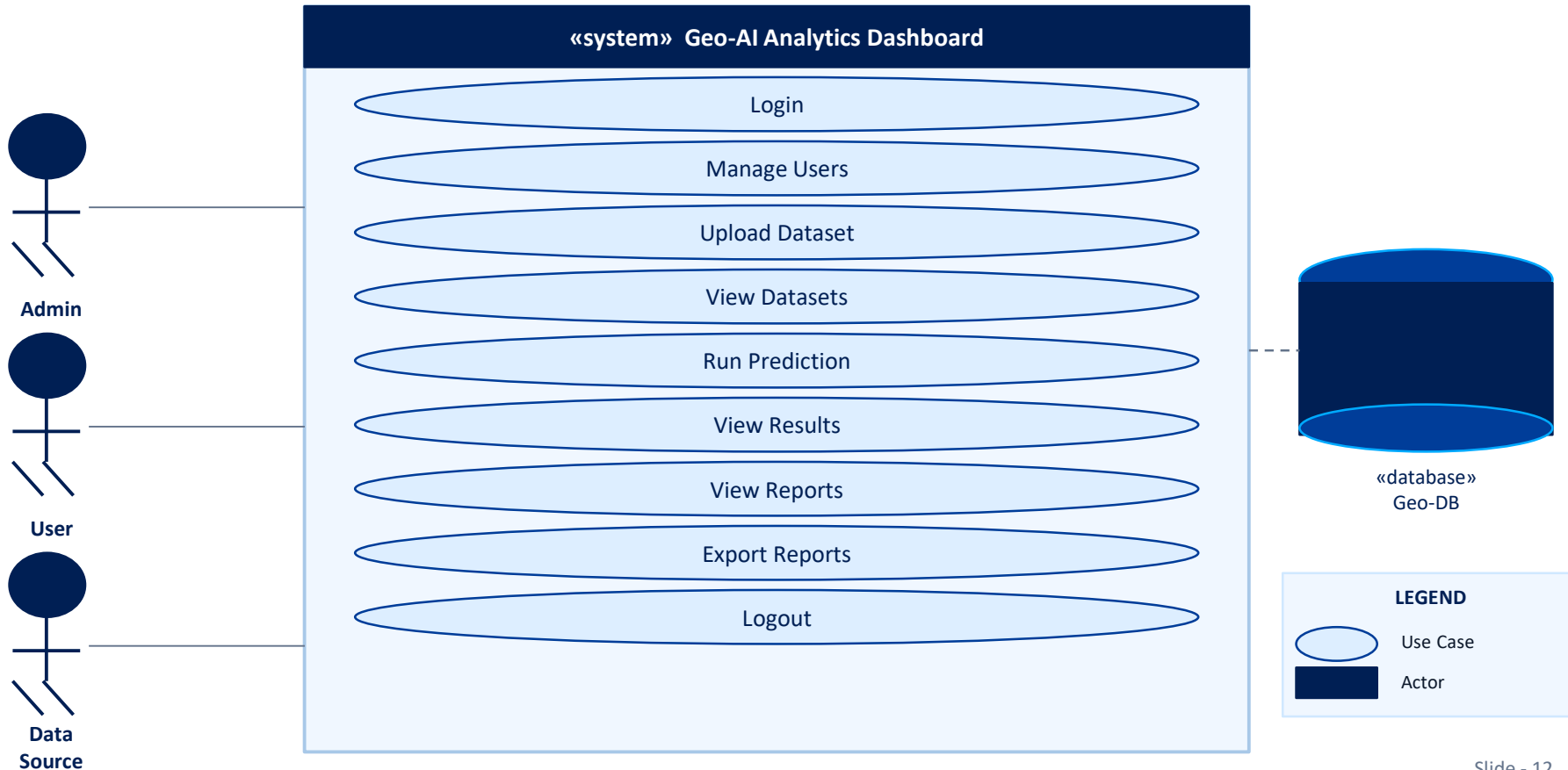
↩ Module Integration: Frontend UI → Flask REST API → Preprocessing → ML Model → Prediction Engine → Visualization Dashboard

Non-Functional Requirements: Performance · Reliability · Usability · Security · Scalability · Compatibility · Maintainability · Portability

SYSTEM WORKFLOW / FLOWCHART

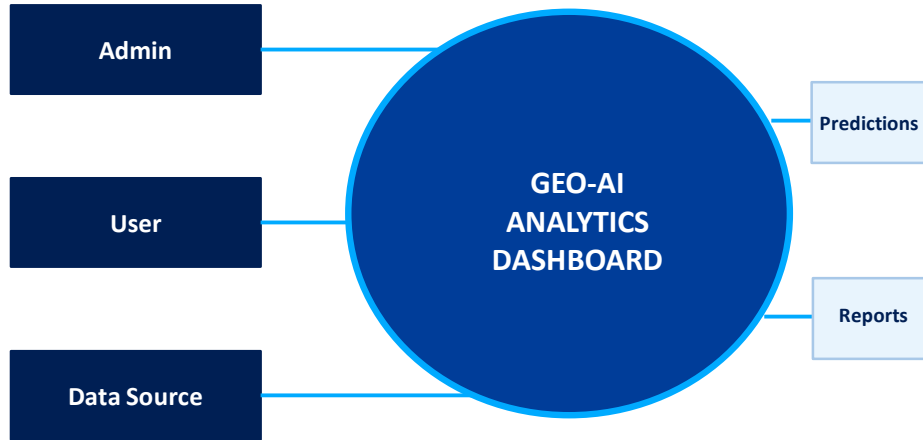


USE CASE DIAGRAM

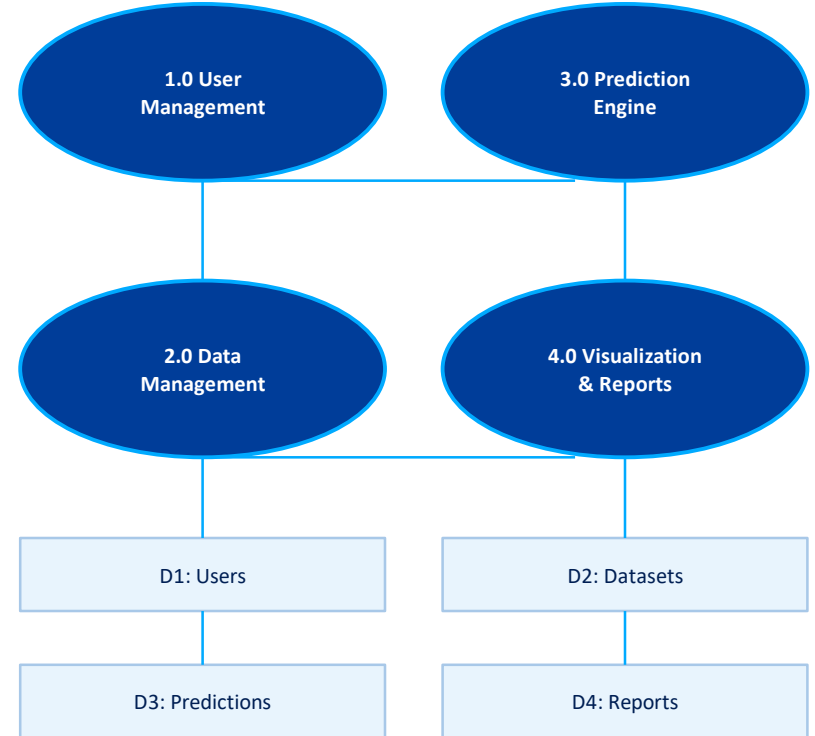


DATA FLOW DIAGRAM (DFD)

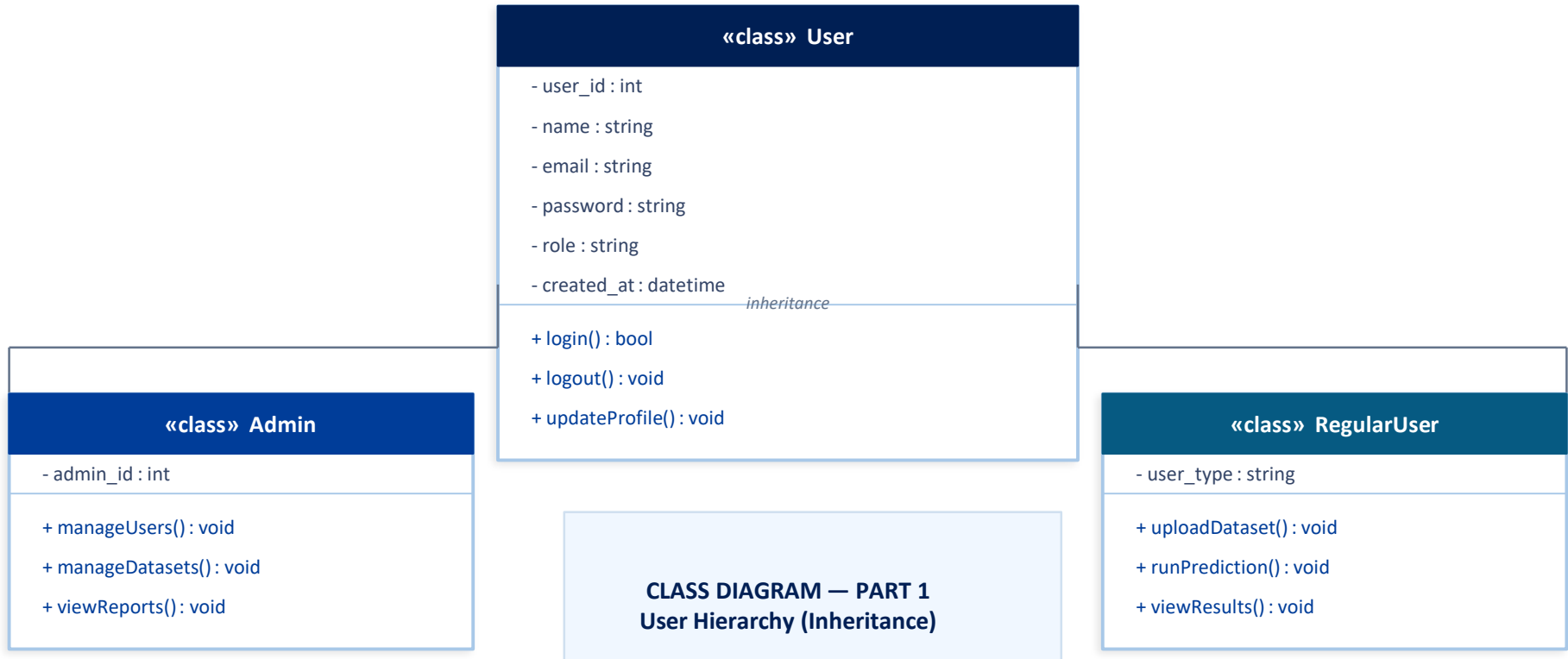
DFD LEVEL 0 — Context Diagram



DFD LEVEL 1 — Subsystem Decomposition



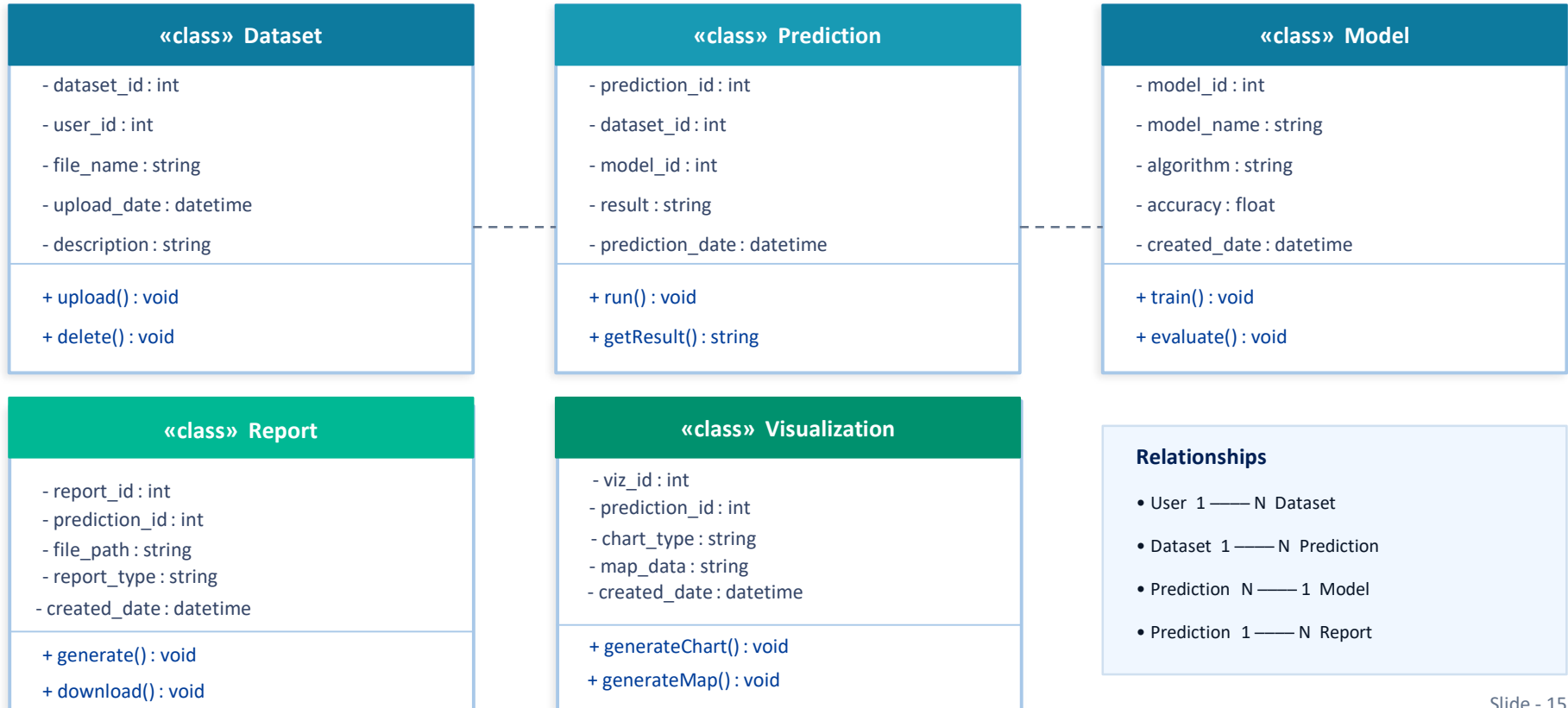
CLASS DIAGRAM



Continued on next slide →

CLASS DIAGRAM — PART 2

← Continued from Part 1



IMPLEMENTATION METHODOLOGY

01

Dataset Preparation

CSV geospatial data with latitude, longitude, rainfall, temperature, vegetation index and label columns.

02

Data Preprocessing

Fill null values → select features → normalize → encode categoricals → generate NDVI → train/test split.

03

ML Model Training

Random Forest Classifier trained using Scikit-learn; model saved as geo_model.pkl via joblib.

04

Flask Backend API

REST endpoints: /register, /login (JWT), /predict (file upload + ML inference), /history, /admin.

05

Frontend Interface

HTML/CSS/JS dashboard with file upload, prediction output panel, Chart.js bar chart and Leaflet.js map view.

06

Integration & Testing

Frontend → Backend → Preprocessing → ML Model → Output. Unit, Integration, System & UAT testing performed.

TESTING

Unit Testing

Preprocessing function & ML model prediction tested independently for correctness.

Integration Testing

Backend Flask API + ML model pipeline tested end-to-end with CSV input.

System Testing

Full workflow: Upload file → Preprocess → Predict → Visualize — tested as complete system.

User Acceptance Testing

Interface usability, responsive dashboard and prediction history verified from user perspective.

Test Cases

Test ID	Input	Expected Output	Result
TC01	Valid CSV file	Predictions generated	✓ Pass
TC02	Missing values	Handled properly	✓ Pass
TC03	Invalid format	Error message shown	✓ Pass
TC04	Large dataset	Processed successfully	✓ Pass

OUTPUT SCREENSHOTS

• • • LOGIN PAGE

Login Page

JWT-secured login with username/password, toggle eye visibility.
Clean gradient UI.

• • • REGISTER PAGE

Register Page

New user registration form with back-to-login navigation.

• • • HOME / UPLOAD DATASET

Home / Upload Dataset

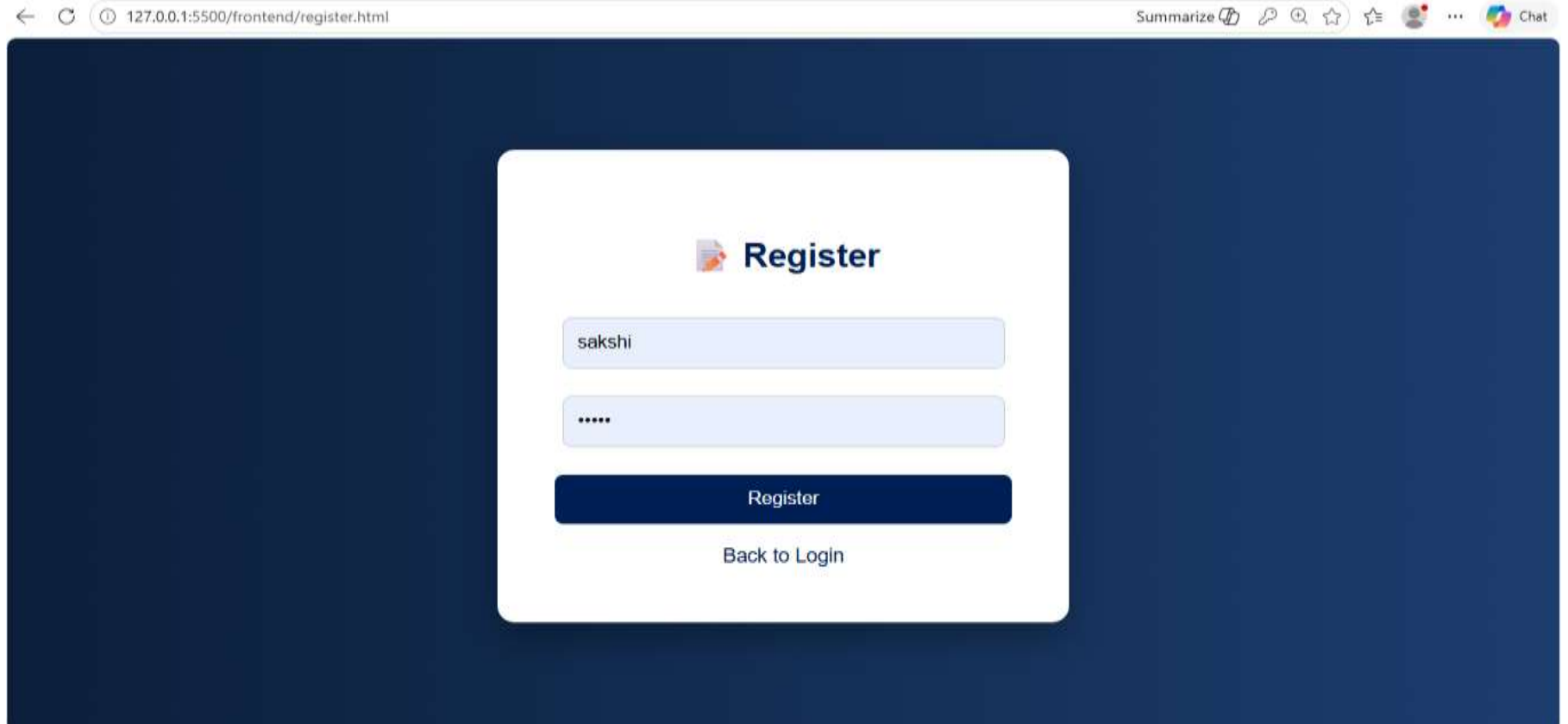
Main dashboard with CSV file upload, Predict / History / Admin / Logout buttons.

• • • DATA VISUALIZATION & MAP

Data Visualization & Map

Prediction output panel, Chart.js bar chart (Safe vs Risk) and live Leaflet.js geospatial map view.

OUTPUT SCREENSHOTS



The screenshot displays a web browser window with the address bar showing "127.0.0.1:5500/frontend/register.html". The browser's top right corner includes navigation icons and a "Chat" button. The main content area features a dark blue background with a central white rounded rectangle. Inside this rectangle, the word "Register" is displayed in a large, bold, dark blue font, preceded by a small icon of a document with a red checkmark. Below the title, there are two light blue input fields. The first field contains the text "sakshi". The second field contains six dots, indicating a password. Below these fields is a dark blue button with the text "Register" in white. At the bottom of the white rectangle, there is a link that says "Back to Login".

127.0.0.1:5500/frontend/register.html

Summarize

Chat

Register

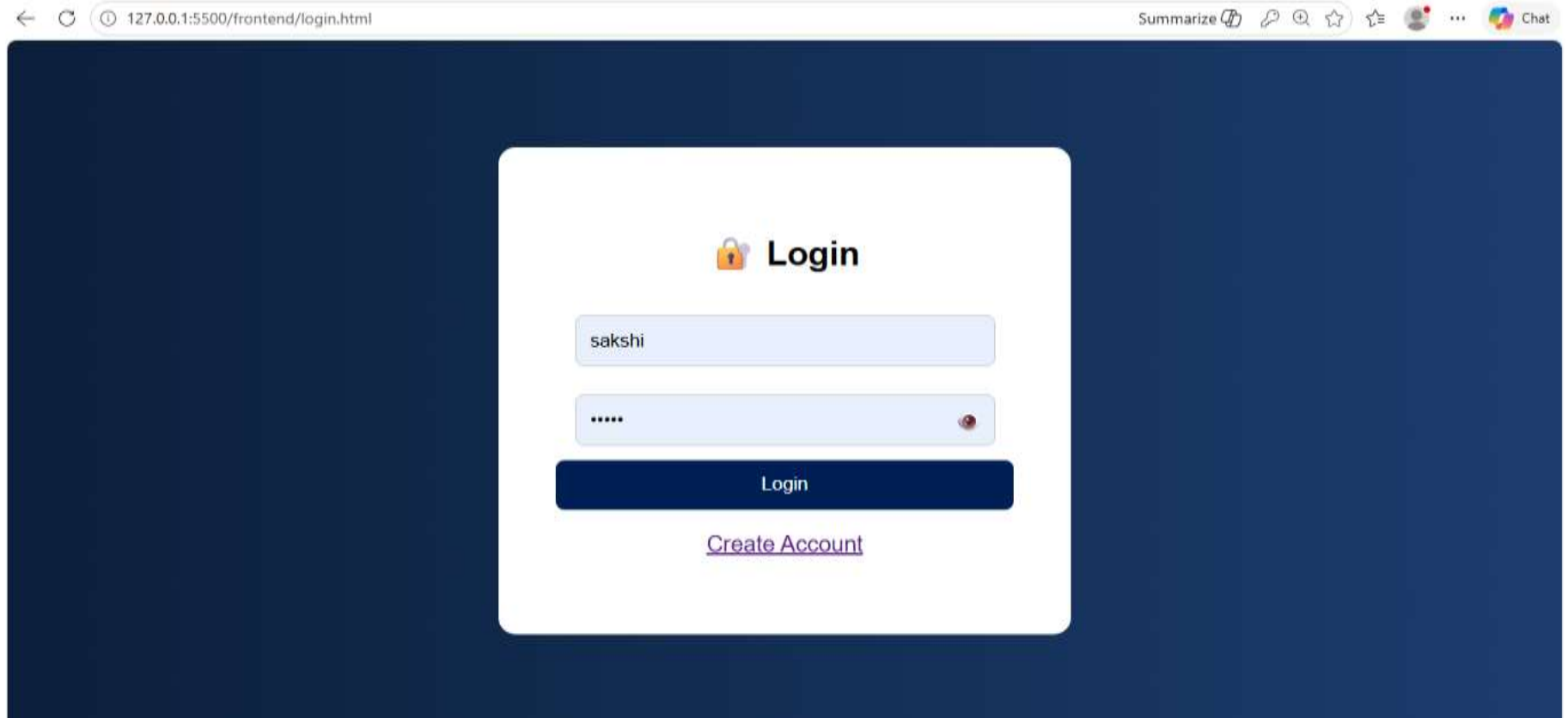
sakshi

.....

Register

[Back to Login](#)

OUTPUT SCREENSHOTS



OUTPUT SCREENSHOTS

The screenshot displays a web application titled "GEO-AI ANALYTICS DASHBOARD | INSPIRED BY SPACE SYSTEMS". The interface is divided into two main sections: "Data Upload" and "Prediction Output".

Data Upload Section:

- Contains a "Choose File" button and the text "No file chosen".
- Below the upload area are four buttons: "Predict", "History", "Admin", and "Logout".

Prediction Output Section:

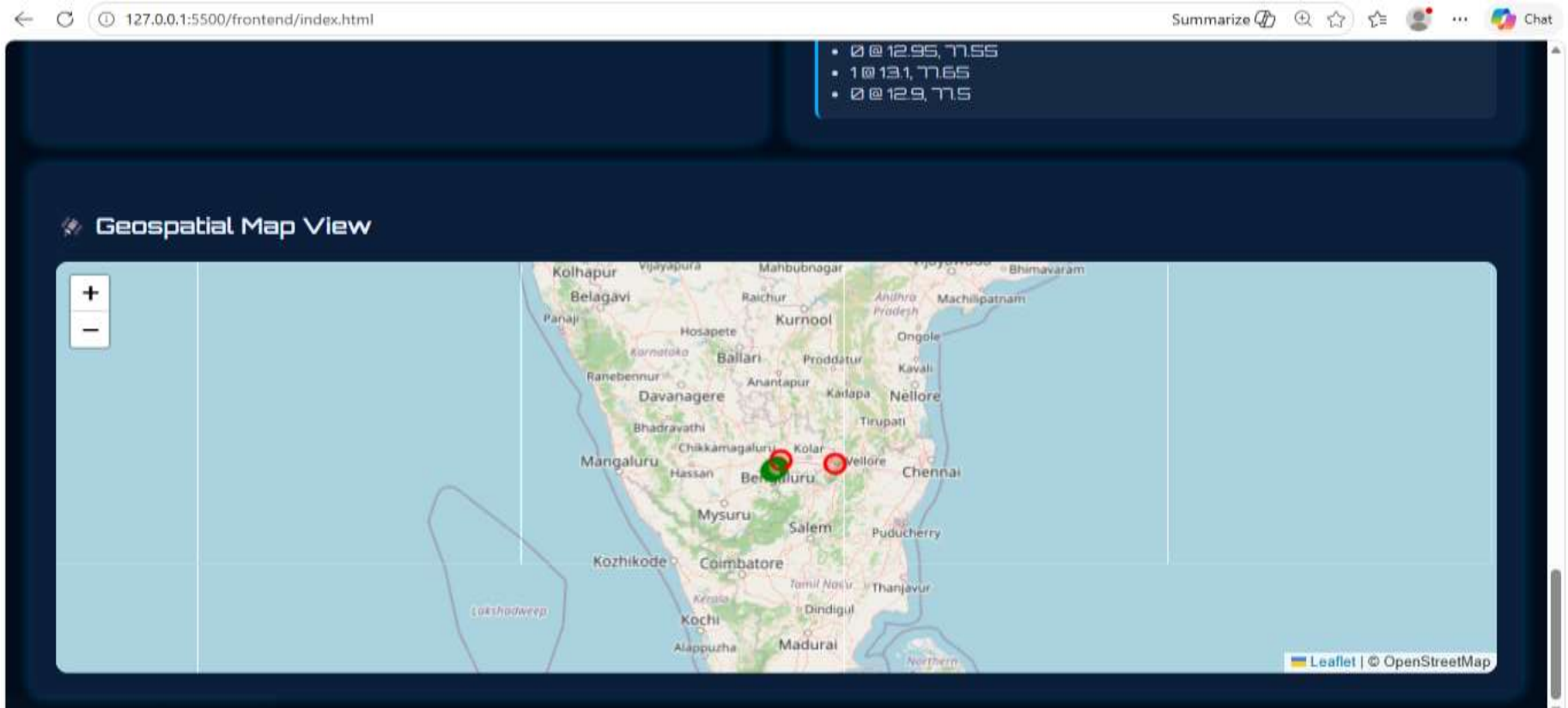
Summary statistics: Total points: 5, Normal: 3, Alert: 2, Accuracy: 0.6

#	Latitude	Longitude	Status
1	12.97	77.59	Normal
2	13.02	78.6	Alert
3	12.95	77.55	Normal
4	13.1	77.65	Alert
5	12.9	77.5	Normal

OUTPUT SCREENSHOTS



OUTPUT SCREENSHOTS



CONCLUSION



Successfully developed an end-to-end Geospatial ML System for environmental risk zone prediction using latitude, longitude, rainfall, temperature and vegetation index.



Integration of web-based interface with ML backend enables seamless dataset upload, real-time prediction and interactive chart & map visualization.



Modular RESTful architecture ensures scalability, flexibility and efficient data flow; JWT authentication provides secure multi-user access.



All test cases passed. Model achieves 85% accuracy on the baseline dataset, demonstrating the viability of ML for geospatial analytics.



Project establishes a strong foundation for geospatial intelligence systems supporting disaster management, agriculture, environmental monitoring and urban planning.

FUTURE ENHANCEMENTS

1. Advanced ML Models

XGBoost, deep learning architectures; large-scale real-world datasets with hyperparameter tuning.

2. Full GIS Integration

QGIS/ArcGIS support; raster & vector data processing; DBSCAN/K-Means spatial clustering.

3. Real-Time Data Feeds

Live APIs for weather, satellite imagery and IoT environmental sensor streams.

4. Explainable AI (XAI)

SHAP feature importance; model interpretability dashboards for transparent decisions.

5. Cloud Deployment

Microservices with Docker; deploy on AWS/Azure/IBM Cloud for enterprise-grade scalability.

6. Security & Governance

Role-based access control; strong password hashing; data encryption & privacy compliance.

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6. Kaggle – Datasets and ML Resources – <https://www.kaggle.com>

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12. TensorFlow/Keras Documentation – <https://www.tensorflow.org>

13. HTML5 / JavaScript – Web Standards – <https://developer.mozilla.org>

THANK YOU

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